

Does model scale improve verb prediction across long dependencies in Japanese?

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Research context & gap

Japanese is a verb-final SOV language: the main verb arrives only after all arguments, often separated by many intervening phrases. Prior work shows English LLMs handle long-distance dependencies better as scale increases, but this has *not* been systematically tested in Japanese, where verb-finality places uniquely high demands on memory across a sentence.

Research question: Does parameter count predict a model’s ability to anticipate the sentence-final verb given increasing structural distance?

Study design

Corpus & metric

3,798 sentences from UD Japanese GSD.
Surprisal = $-\log P$, summed over verb tokens.
Distance = bunsetsu heads between first case-marked argument and root verb.
Bunsetsu = smallest functional unit of a Japanese sentence

Controls

OLS regression controls for log sentence length and log verb frequency.
Log-transform on distance addresses right skew.
Null sentences excluded.

UD Japanese GSD corpus summary

3,798

sentences analyzed

10.99

mean bunsetsu distance

2 – 74

distance range

0.775

distance $\times$ length corr.

Results by model

Small · ~340M params

rinna/japanese-gpt2-medium

Mean surprisal: 4.11 (SD 2.83)

Distance β = +0.145 ($p = 0.033^*$)

Short vs. long diff: +0.534

Well-behaved

Medium · ~1.3B params

rinna/japanese-gpt-1b

Mean surprisal: 6.95 (SD 5.45)

Distance β = +0.181 ($p = 0.163$ n.s.)

Short vs. long diff: -0.215

Anomalous

Large · ~3B params

cyberagent/open-calm-3b

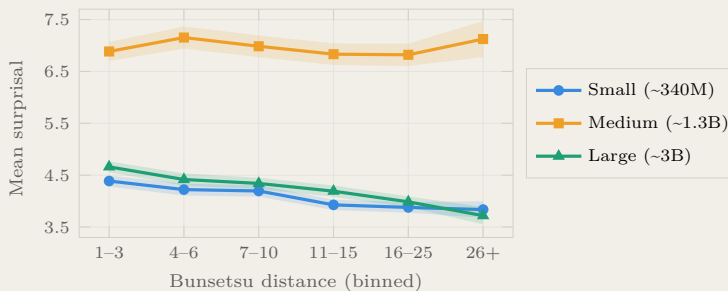
Mean surprisal: 4.29 (SD 2.90)

Distance β = +0.193 ($p = 0.005^{**}$)

Short vs. long diff: +0.744

Well-behaved

Mean verb surprisal by distance bin ($\pm$ SE)



Small and large show a consistent downward trend across distance bins; medium remains elevated and flat throughout.

Crucially, the large model does not outperform the small, suggesting parameter count alone does not improve verb prediction in Japanese sentences with long dependencies.

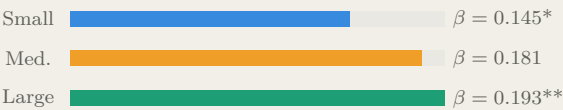
This visual pattern is corroborated by the controlled regression: all three models return positive `log_dist.` coefficients, confirming the distance effect persists regardless of scale.

Full regression summary

	Sm.	Med.	Lg.
Predictor			
Intercept	7.41***	7.64***	8.58***
log_dist.	0.145*	0.181	0.193**
log_len.	-0.87***	-0.88***	-1.22***
log_freq.	-0.31***	+0.76***	-0.29***
R^2	0.059	0.074	0.066
N	3,798		

* $p < .05$ ** $p < .01$ *** $p < .001$

Distance coefficient (`log_dist.` β)



All $\beta > 0$: distance increases controlled surprisal regardless of the model’s scale, contrary to English.

Discussion & gap addressed

What this project shows

Contrary to the English-centric finding that larger models better tolerate long dependencies, scaling from 340M to 3B parameters does *not* reduce verb surprisal or dampen the distance effect in Japanese. The raw trend, longer distance $\rightarrow$ lower surprisal, reflects a corpus-level confound: longer sentences tend to contain more predictable, high-frequency verbs. Once controlled for, the distance effect reverses, and the large model’s coefficient (0.193) is actually *larger* than the small model’s (0.145), with near-identical mean surprisal (~4.1 vs. ~4.3).

Why this matters

Japanese verb-finality may place qualitatively different demands on context integration than English, ones that parameter count alone does not address in the 340M–3B range. The medium model’s anomalous behavior (mean surprisal ≈ 7 , non-significant distance effect, inverted frequency coefficient) could be attributed to tokenization artifacts in the rinna-1B vocabulary, not a genuine scale effect, shown by similar behavior in the small and large models. Future work should test instruction-tuned or larger-scale ($>7B$) Japanese models.